

BUSFIN 8200: Asset Pricing Foundations

Problem Set 1

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1 Question 1

1.1 Derivation of Equations 1.1–1.3

Throughout, a lowercase letter denotes the log of the corresponding uppercase variable, e.g. $r_{t+1} = \log(R_{t+1})$, $p_t = \log(P_t)$, and $cf_{t+1} = \log(CF_{t+1})$.

Step 1: log return. Start from the definition of the gross return,

$$R_{t+1} = \frac{P_{t+1} + CF_{t+1}}{P_t}. \quad (1)$$

Taking logs,

$$\begin{aligned} \log(R_{t+1}) &= \log(P_{t+1}) - \log(P_t) + \log\left(1 + \frac{CF_{t+1}}{P_{t+1}}\right) \\ \iff r_{t+1} &= p_{t+1} - p_t + \log\left(1 + e^{cf_{t+1} - p_{t+1}}\right). \end{aligned} \quad (2)$$

Step 2: first-order Taylor approximation of the log term. Rearranging (2) so that the polynomial terms are on the left-hand side and the log term is on the right-hand side,

$$r_{t+1} - p_{t+1} + p_t = \log\left(1 + e^{cf_{t+1} - p_{t+1}}\right). \quad (3)$$

Let $x_{t+1} = cf_{t+1} - p_{t+1}$. The function $\log(1 + e^x)$ can be approximated by a first-order Taylor expansion around a point a :

$$\log(1 + e^x) \approx \log(1 + e^a) + \frac{e^a}{1 + e^a} (x - a), \quad (4)$$

which is accurate for small enough $x - a$. The expansion point a is taken to be the time average of $cf_{t+1} - p_{t+1}$, which is assumed to converge to a constant, denoted $a = \overline{cf} - \bar{p}$.

Step 3: substituting the approximation. Applying the Taylor approximation (4) to the right-hand side of (3), with $x = cf_{t+1} - p_{t+1}$ and $a = \overline{cf} - \bar{p}$, and rearranging the terms, we obtain

$$p_t = \kappa_0 + (1 - \kappa) cf_{t+1} - r_{t+1} + \kappa p_{t+1}, \quad (5)$$

where, as in the problem statement,

$$\kappa = \frac{1}{1 + e^a} \quad \text{and} \quad \kappa_0 = -\log(\kappa) - (1 - \kappa) \log\left(\frac{1}{\kappa} - 1\right), \quad (6)$$

with $a = \overline{cf} - \bar{p}$ the expansion point of Step 2 (the problem statement writes this average as $\overline{dp}$).

Step 4: rewriting in terms of the log cash-flow–price ratio. Subtracting cf_t from both sides of (5) and multiplying both sides by -1 , and defining

$$dp_t := cf_t - p_t \quad \text{and} \quad \Delta cf_{t+1} := cf_{t+1} - cf_t, \quad (7)$$

we obtain

$$dp_t = -\kappa_0 + \kappa dp_{t+1} + r_{t+1} - \Delta cf_{t+1}. \quad (8)$$

Step 5: forward substitution. Equation (8) holds at every date, so shifting it one period forward,

$$dp_{t+1} = -\kappa_0 + \kappa dp_{t+2} + r_{t+2} - \Delta cf_{t+2}. \quad (9)$$

Plugging (9) into the dp_{t+1} term on the right-hand side of (8) gives

$$dp_t = -\kappa_0 - \kappa \kappa_0 + \kappa^2 dp_{t+2} + r_{t+1} + \kappa r_{t+2} - \Delta cf_{t+1} - \kappa \Delta cf_{t+2}. \quad (10)$$

Step 6: substitution up to horizon H . Continuing to substitute the future dp terms into (10) up to horizon H , we get

$$dp_t = -\kappa_0 \sum_{h=0}^{H-1} \kappa^h + \kappa^H dp_{t+H} + \sum_{h=1}^H \kappa^{h-1} r_{t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta cf_{t+h}. \quad (11)$$

1.2 Variance decomposition by horizon (data)

1.3 Variance decomposition by horizon (VAR-implied)

1.4 Infinite-horizon decomposition

1.5 Derivation of Equations 1.7–1.9 (Gao and Martin, 2021)

2 Question 2

2.1 Predictive regressions by horizon

Figure 1 plots the adjusted R^2 of the regression

$$\frac{1}{H} \sum_{h=1}^H xR_{e,t+h} = a^{(H)} + b^{(H)} \frac{D_t}{P_t} + \varepsilon_t^{(H)}, \quad H = 1, \dots, 15, \quad (12)$$

estimated by OLS on the EQ Dataset, where $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{dp_t}$. The data are monthly observations of annual variables, so $t+h$ is $12h$ months after t , and each horizon uses every month t for which H years of subsequent data are available (code: `src/q2a.py`).

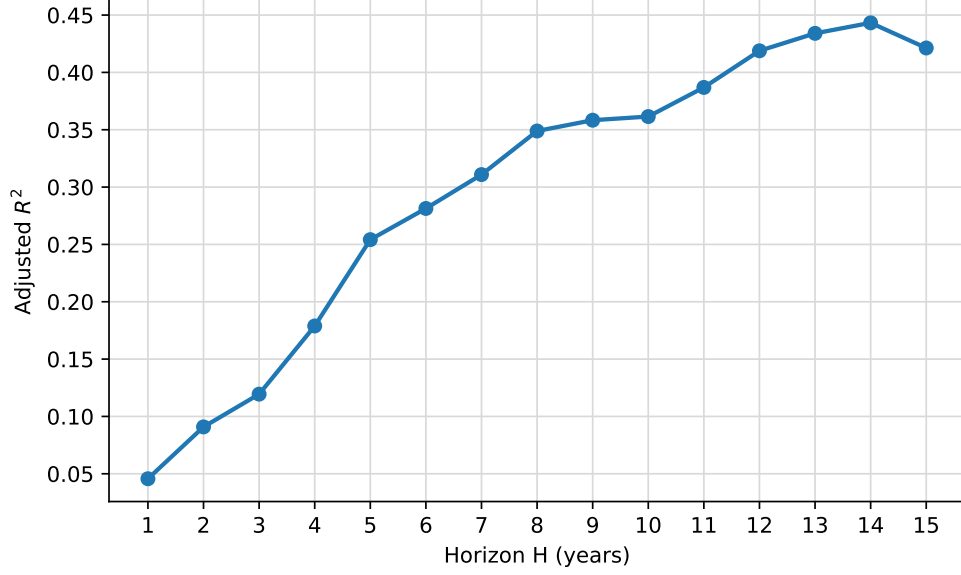


Figure 1: Adjusted R^2 of regression (12) against the horizon H (in years). Monthly observations, 1927:12–2021:12.

2.2 Standard errors for the one-year predictive regression

Table 1 reports the OLS estimates of

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + \varepsilon_t, \quad (13)$$

together with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. As in Question 2a, $xR_{e,t} = e^{r_{e,t}} - e^{r_{f,t}}$ and $D_t/P_t = e^{d_{p,t}}$; $t+1$ is 12 months after t , and the sample uses every month t for which the subsequent annual return is available (code: `src/q2b.py`).

Standard-error method	SE($\hat{b}$)	$t(\hat{b})$	Lags
(i) OLS	0.3802	7.38	–
(ii) White (1980)	0.6587	4.26	–
(iii) Newey-West (1987), 11 lags	1.2979	2.16	11
(iv) Hansen-Hodrick (1980), 11 lags	1.4521	1.93	11
(v) Newey-West (1987, 1994), automatic lags	1.2311	2.28	24
$\hat{a} = -0.0314$, $\hat{b} = 2.8038$, $T = 1117$ monthly observations			

Table 1: OLS estimates of regression (13) with the standard error and t -statistic of $\hat{b}$ under five standard-error methods. Predictor dates 1927:12–2020:12. “Lags” is the number of autocorrelation lags; for method (v) it is selected by the Newey–West (1994) automatic bandwidth rule.

2.3 Amihud and Hurvich (2004) bias-corrected estimate

Table 2 reports the Amihud and Hurvich (2004) estimate $\hat{b}^c$ of b in regression (13), on the same sample as Table 1. First, D_{t+1}/P_{t+1} is regressed on D_t/P_t by OLS, giving $\hat{\theta}$ and $\hat{\rho}$, and the

slope is bias-corrected as $\hat{\rho}^c = \hat{\rho} + (1 + 3\hat{\rho})/n + 3(1 + 3\hat{\rho})/n^2$ with $n = 94$ years. Second, with $\hat{u}_{t+1}^c = D_{t+1}/P_{t+1} - (\hat{\theta} + \hat{\rho}^c D_t/P_t)$,

$$xR_{e,t+1} = a + b \frac{D_t}{P_t} + b_u \hat{u}_{t+1}^c + \varepsilon_t \quad (14)$$

is estimated by OLS, and $\hat{b}^c$ is the coefficient on D_t/P_t . Its standard error follows Amihud and Hurvich (2004, Eq. 10), $\widehat{SE}(\hat{b}^c) = \sqrt{(\hat{b}_u)^2 (1 + 3/n + 9/n^2)^2 \widehat{\text{Var}}(\hat{\rho}) + \widehat{SE}(\hat{b}^c)^2}$ with $n = 94$ (code: `src/q2c.py`).

Standard error of $\hat{b}^c$	$\text{SE}(\hat{b}^c)$	$t(\hat{b}^c)$
(i-a) Eq. (10), OLS ingredients	0.3876	6.03
(i-b) Eq. (10), Newey-West (1987) 11 lags ingredients	1.5170	1.54
$\hat{b}^c = 2.3362, \quad \hat{a} = -0.0314, \quad \hat{b}_u = -13.4837$		
$\hat{\rho} = 0.7197, \quad \hat{\rho}^c = 0.7544, \quad n = 94 \text{ years}, \quad T = 1117 \text{ monthly observations}$		
OLS $\hat{b}$ (Question 2b) = 2.8038		

Table 2: Amihud and Hurvich (2004) estimate of b in regression (13) via the augmented regression (14). In row (i-a), $\widehat{\text{Var}}(\hat{\rho})$ and $\widehat{SE}(\hat{b}^c)$ are plain OLS estimates; in row (i-b), both are Newey–West (1987) estimates with 11 lags. Predictor dates 1927:12–2020:12.

2.4 Out-of-sample estimation (expanding window)

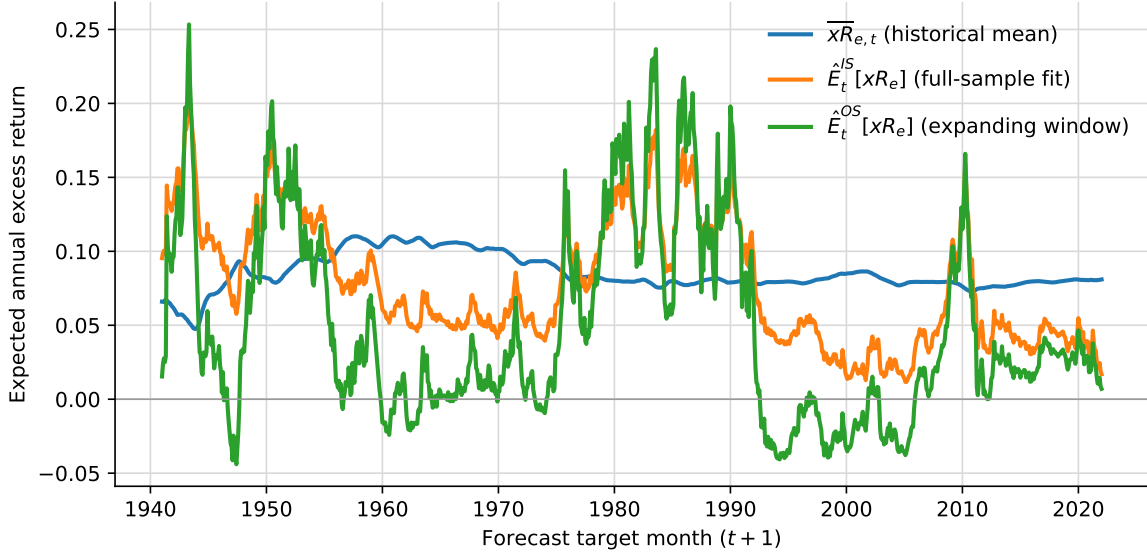


Figure 2: Expanding-window historical mean $\overline{xR}_{e,t}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e]$ from regression (13) estimated with data available at t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

The out-of-sample R^2 over the forecast-target months 1940:12–2021:12 is $R_{OS}^2 = -0.0135$.

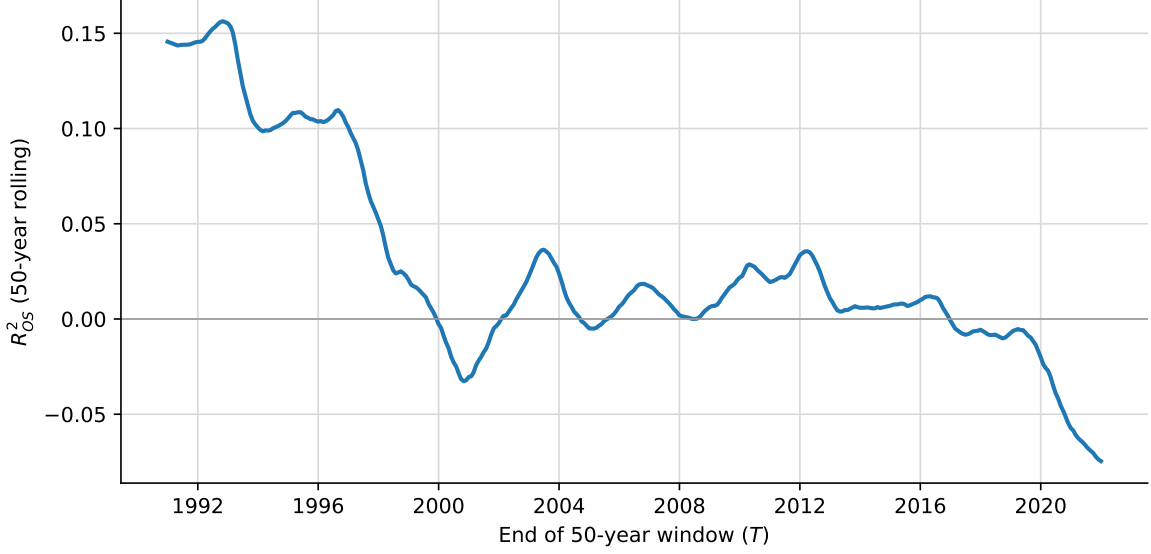


Figure 3: R^2_{OS} computed over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, with $\hat{a}_t$, $\hat{b}_t$ and $\overline{xR}_{e,t}$ unchanged.

2.5 Out-of-sample estimation with steady-state restrictions

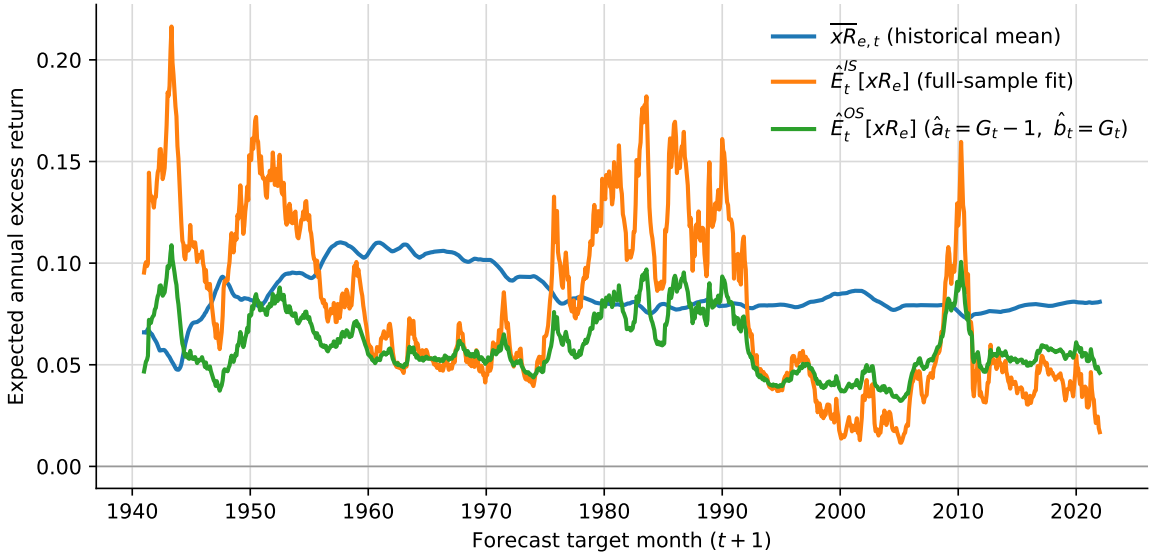


Figure 4: Expanding-window historical mean $\overline{xR}_{e,t}$, in-sample fitted values $\hat{E}_t^{IS}[xR_e]$ from the full-sample regression of Question 2b, and out-of-sample forecasts $\hat{E}_t^{OS}[xR_e] = (G_t - 1) + G_t D_t/P_t$, where G_t is the expanding mean of $e^{\Delta d}$ over 1927:12 to t , plotted against the forecast-target month $t + 1$, 1940:12–2021:12.

The out-of-sample R^2 over the forecast-target months 1940:12–2021:12 is $R^2_{OS} = 0.0294$.

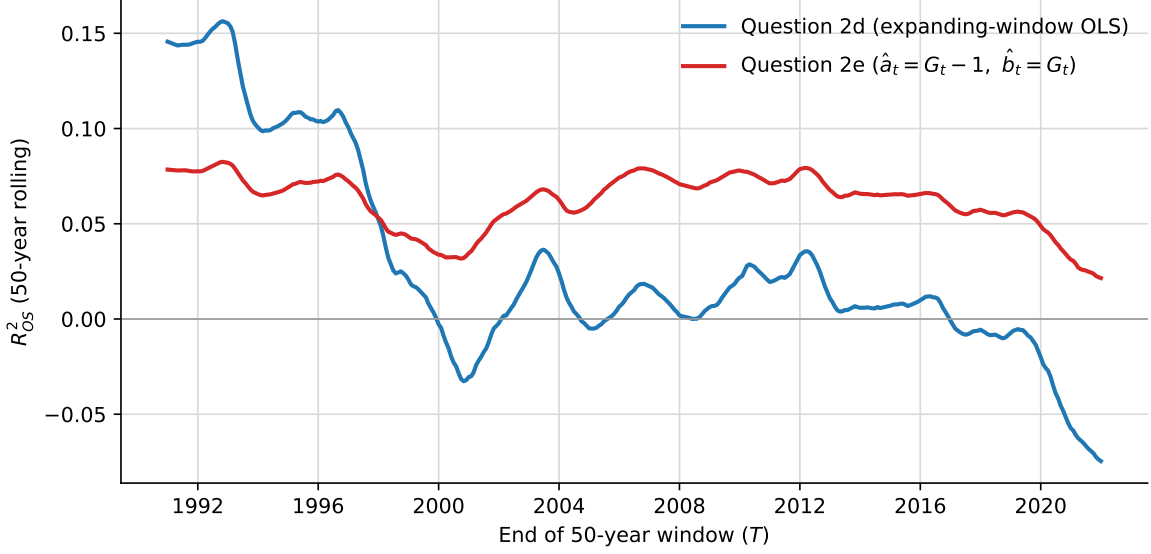


Figure 5: R_{OS}^2 over a rolling window of 600 forecast-target months ending at T , for T from 1990:12 to 2021:12, for the expanding-window OLS forecasts of Question 2d and the restricted forecasts of Question 2e ($\hat{a}_t = G_t - 1$, $\hat{b}_t = G_t$).

3 Question 3

3.1 Momentum signal construction and validation

3.2 Book-to-market signal construction and validation

3.3 Decile portfolios

3.4 Fama–MacBeth regressions

3.5 Portfolio-level multivariate regressions

4 Question 4

4.1 Yields, forward rates, and returns

From the Fama–Bliss yields $Y_{b,t}^{(H)}$ (in percent) for $H = 1, \dots, 5$, the log yields, log forward rates and log annual returns are

$$y_{b,t}^{(H)} = \log\left(1 + Y_{b,t}^{(H)}/100\right), \quad f_{b,t}^{(H)} = H y_{b,t}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad r_{b,t}^{(H)} = H y_{b,t-1}^{(H)} - (H-1) y_{b,t}^{(H-1)}, \quad (15)$$

with $y^{(0)} = 0$, so that $f_{b,t}^{(1)} = y_{b,t}^{(1)}$ and $r_{b,t}^{(1)} = y_{b,t-1}^{(1)}$; $t-1$ is 12 months before t . The spreads are $xy^{(H)} = y^{(H)} - y^{(1)}$, $xf^{(H)} = f^{(H)} - f^{(1)}$ and $xr^{(H)} = r^{(H)} - r^{(1)}$ (code: `src/q4a.py`).

Average (%)	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$xy_{b,t}^{(H)}$ (yield spread)	0.169	0.331	0.471	0.568
$xf_{b,t}^{(H)}$ (forward spread)	0.339	0.654	0.892	0.956
$xr_{b,t}^{(H)}$ (excess return)	0.315	0.607	0.820	0.872
Common sample 1953:06–2024:12				

Table 3: Average yield spreads, forward spreads and excess log returns of H -year Fama–Bliss discount bonds over the 1-year bond, $H = 2, \dots, 5$, in percent per year. Monthly observations over the common sample on which all three series are defined, 1953:06–2024:12.

4.2 Hold-to-maturity excess return regressions

Table 4 reports OLS estimates of

$$\frac{1}{H} xr_{b,t \rightarrow t+H}^{(H)} = a^{(H)} + b^{(H)} xy_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (16)$$

where the hold-to-maturity excess return is

$$xr_{b,t \rightarrow t+H}^{(H)} = \sum_{h=1}^H xr_{b,t+h}^{(H-h+1)} = \sum_{h=1}^H \left(r_{b,t+h}^{(H-h+1)} - r_{b,t+h}^{(1)} \right), \quad (17)$$

with r and xr as in (15), $t+h$ being $12h$ months after t , and the $h = H$ term identically zero because $xr_{b,t}^{(1)} = 0$. Each regression uses every month t for which all non-zero terms of (17) are observed, i.e. $t + 12(H-1) \leq 2024:12$. The t -statistics use Hansen–Hodrick (1980) standard errors with $L = 12H - 1$ lags (code: `src/q4b.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.532	0.414	0.345
	[3.56]	[3.14]	[2.46]	[2.25]
R^2 (%)	7.7	5.2	3.8	3.2

Table 4: OLS estimates of $b^{(H)}$ in regression (16), with Hansen–Hodrick (1980) t -statistics ($12H - 1$ lags) in brackets and R^2 in percent. Monthly observations from 1952:06; the last predictor month is 2023:12, 2022:12, 2021:12 and 2020:12 for $H = 2, 3, 4, 5$ (859, 847, 835 and 823 observations).

4.3 One-year excess return regressions on forward spreads

Table 5 reports OLS estimates of

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} xf_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad H = 2, \dots, 5, \quad (18)$$

with xr and xf as in (15) and $t+1$ being 12 months after t . Each regression uses every month t for which $xr_{b,t+1}^{(H)}$ is observed, i.e. $t + 12 \leq 2024:12$, so all four samples are 1952:06–2023:12. The t -statistics use Newey–West (1987) standard errors with the lag length selected by the Newey–West (1994) automatic bandwidth rule, as in method (v) of Question 2b (code: `src/q4c.py`).

	$H = 2$	$H = 3$	$H = 4$	$H = 5$
$b^{(H)}$	0.677	0.889	1.116	0.964
	[3.23]	[3.35]	[3.63]	[2.92]
R^2 (%)	7.7	8.6	10.7	6.7

Table 5: OLS estimates of $b^{(H)}$ in regression (18), with Newey–West (1987, 1994) t -statistics in brackets (selected lags 23, 23, 24 and 23 for $H = 2, 3, 4, 5$) and R^2 in percent. Monthly observations, predictor months 1952:06–2023:12 (859 observations for each H).

4.4 Cochrane–Piazzesi factor

4.5 Excess return regressions on the CP factor